

Track 2D h0 Offset Cross-Check

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

Cross-check of Track 2D signed model offset errors against measured h_0 / curve mean over 12416 joined candidate-curve rows and 111 candidates.

Decision

- h_0 magnitude alone does not explain most Track 2D offset failures.
- Median candidate $\text{abs}(\text{error})$ versus $\text{abs}(h_0)$ correlation is -0.0152 ; mean is -0.0125 .
- Median top-decile overlap lift is 1.0000 versus a random-decile baseline of 1.0 ; 33 of 111 candidates reach lift ≥ 2.0 .
- 94 of 111 candidates have weak absolute correlation < 0.25 .
- Join validation is tight: maximum $\text{Track 2D truth_mean_deg} - \text{measured_h0_deg}$ is 0.0000 deg.

Surface Summary

Surface	Rows	Mean Abs Offset	Mean Abs h0	Abs Corr	Overlap Lift	Max Delta
Bw	3977	0.0044	0.0263	-0.0885	1.4824	0.0000
Fw	5141	0.0032	0.0585	0.0181	1.7476	0.0000
global	3298	0.0039	0.0424	-0.0044	1.0303	0.0000

Largest Mean Offset Error Candidates

Candidate	Rows	Mean Abs Offset	Mean Abs h0	Abs Corr	Overlap Lift
MLP19_Fw	97	0.0185	0.0585	-0.0388	1.0000
harmonic_regression_global	194	0.0180	0.0424	0.0744	1.5000
MLP19_Bw	97	0.0137	0.0263	-0.1186	2.0000
rcim_retuned_XGBM19_Bw	97	0.0101	0.0263	0.3105	5.0000
rcim_original_MLP19_Fw	97	0.0098	0.0585	-0.0789	0.0000
ELM19_Bw	97	0.0097	0.0263	-0.0313	3.0000
rcim_retuned_MLP19_Bw	97	0.0090	0.0263	-0.0431	1.0000
rcim_retuned_MLP19_Fw	97	0.0083	0.0585	-0.0620	2.0000
rcim_retuned_ELM19_Bw	97	0.0081	0.0263	0.3107	4.0000
XGBM19_Bw	97	0.0077	0.0263	-0.2231	0.0000

Strongest h0/Error Overlap Candidates

Candidate	Rows	Abs Corr	High Error	High h0	Overlap	Lift
rcim_retuned_LGBM19_Bw	97	0.3199	10	10	6	6.0000
rcim_retuned_XGBM19_Bw	97	0.3105	10	10	5	5.0000
rcim_retuned_HGBM19_Bw	97	0.2870	10	10	5	5.0000
LGBM19_Fw	97	0.0258	10	10	5	5.0000
rcim_retuned_ELM19_Bw	97	0.3107	10	10	4	4.0000
XGBM19_Fw	97	0.1168	10	10	4	4.0000
ELM19_Fw	97	0.3372	10	10	3	3.0000
temporal_convolution_Bw	97	0.0720	10	10	3	3.0000
GBM19_Fw	97	0.0508	10	10	3	3.0000
residual_harmonic_lstm_sequence_sparse_rcim_Fw	97	0.0458	10	10	3	3.0000

Strongest Absolute Correlation Candidates

Candidate	Rows	Abs Corr	Signed Corr	Mean Abs Offset	Mean Abs h0
rcim_retuned_ET19_Fw	97	0.3722	0.0846	0.0017	0.0585
ELM19_Fw	97	0.3372	-0.3204	0.0071	0.0585
rcim_retuned_LGBM19_Bw	97	0.3199	-0.9679	0.0073	0.0263
rcim_retuned_ELM19_Bw	97	0.3107	-0.5644	0.0081	0.0263
rcim_retuned_XGBM19_Bw	97	0.3105	-0.9868	0.0101	0.0263
rcim_retuned_HGBM19_Bw	97	0.2870	-0.7289	0.0034	0.0263
rcim_original_ELM19_Fw	97	0.2581	-0.3404	0.0050	0.0585
rcim_retuned_RF19_Fw	97	0.1682	0.0298	0.0012	0.0585
rcim_original_RF19_Fw	97	0.1618	0.0593	0.0015	0.0585
rcim_original_XGBM19_Fw	97	0.1571	-0.0270	0.0024	0.0585

High-Error Normal-h0 Quadrants

These rows are important because they contradict a pure `h0` -magnitude explanation.

Candidate	High Error + High h0	High Error + Normal h0	Normal Error + High h0
feedforward_global	0	20	20
periodic_temporal_convolution_global	1	19	19
residual_harmonic_gru_sequence_dense240_global	1	19	19
residual_harmonic_lstm_sequence_dense240_global	1	19	19
temporal_convolution_global	1	19	19
gru_sequence_global	2	18	18
periodic_gru_sequence_global	2	18	18
residual_harmonic_gru_sequence_dense360_global	2	18	18
residual_harmonic_gru_sequence_sparse_rcim_global	2	18	18
residual_harmonic_lstm_sequence_dense360_global	2	18	18

Interpretation

The cross-check supports a narrower framing: the problematic quantity is still the curve mean / h_0 channel, but the large Track 2D model offset errors do not simply occur where measured $\text{abs}(h_0)$ is large. This points toward candidate-specific mean prediction bias, direction/regime dependence, or missing causal state information rather than a pure measured- h_0 outlier problem. The useful next diagnostic is therefore predicted-mean versus measured- h_0 surface analysis for the high-error candidates, with separate `Fw` , `Bw` , and `global` handling.

Scatter Diagnostic



Machine-Readable Artifacts

- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_joined_rows.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_candidate_summary.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_surface_summary.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_quadrant_summary.csv`
- `output/validation_checks/track2d_h0_offset_crosscheck/2026-06-09-20-09-16__track2d_h0_offset_crosscheck/track2d_h0_offset_crosscheck_summary.yaml`

Reproduction

```
powershell conda run -n pinns_env python -B  
scripts/reports/analysis/build_track2d_h0_offset_crosscheck_report.py
```